

Chapter 4. Numerical Methods in FE

Financial Engineering

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Content

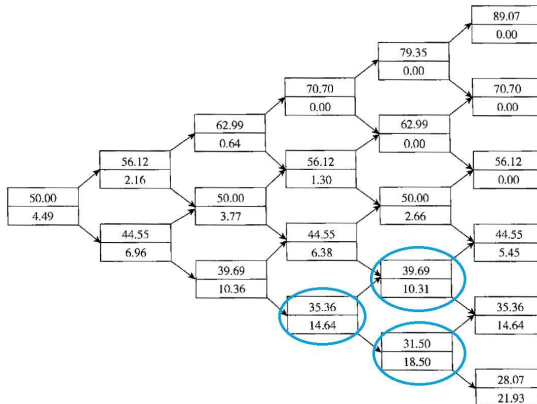
- Discretization (离散化) Techniques
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 - Finite Difference on PDE
- Monti-Carlo Simulation and Its Variance Reduction
- Frontier Methods*
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 - Asymptotic Expansion (e.g., Malliavin Calculus)*

Numerical Schemes in FE

- Financial quantities such as derivative prices are highly complicated, and often do not have any closed-form expression. This problem widely exists in applied mathematics and engineering.
- Designing appropriate numerical algorithm to approximate the true values of financial quantities is a fundamental and significant goal of the scientific research of FE.
- The numerical algorithms are anticipated to be accurate, fast, stable, robust, theoretically convergent, and easy to implement (实现).

Multi-period Binary Tree

The figure below demonstrates the option pricing procedure under the BSM via a multi-period binary tree.



Explanations

- Discretize the time interval into M grid points, where the i -th time point has i nodes (结点). Each node is labeled with two prices for the underlying asset S and the option V , respectively.
- For each price of S , its two leaf nodes have two different new prices: uS and dS , where

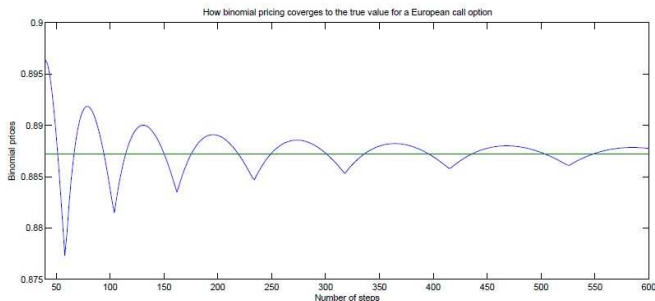
$$u = e^{\sigma\sqrt{\Delta t}}, \quad \text{and} \quad d = 1/u.$$

- Upper probability: $p = \frac{e^{r\Delta t} - d}{u - d}$; down probability $1 - p$
- Risk-neutral recursion (递推)**: the option price on each node is a discounted weighted average:

$$C_{i,j} = e^{-r\Delta t} [pC_{i+1,j} + (1-p)C_{i+1,j+1}].$$

Convergence Effect

An experiments shows that as the grid becomes dense, the approximated results output by the binary trees converge to the true value oscillatedly (振荡地).



Comments

- Simple, but imply the ideas of martingale pricing and risk-neutral measure
- Can deal with American options and exotic options
- Suitable for discretely monitored exotic options
- Difficult for non-BS models
- Convergent, although slowly

Finite Difference Method

- Core idea: discretization. Treat a differential equation (微分方程) as a difference equation (差分方程).
- In various applied disciplines, once a result can be described by the solution to an ODE or PDE, then we often admit that this problem has been solved, since the numerical algorithms on solving ODE and PDE have been highly developed.
- Theoretic tool: Feymann-Kac Theorem in Chapter 2, which expresses the option price as the solution to a parabolic PDE of the second order.
- Input the terminal payoff, and exercise a backward recursion to the initial price.

Detail Procedure (under the BSM)

- Discretization: divide the time and the price into an $N \times M$ grid, and let

$$C_{i,j} := C(i\Delta t, j\Delta x), \quad \text{where } \Delta t := \frac{T}{M}, \quad \Delta x := \frac{x_{\max}}{N}.$$

- Change the PDE of $C(t, x)$ into a difference equation:

$$\frac{C_{i+1,j} - C_{i,j}}{\Delta t} + \left(r - q - \frac{\sigma^2}{2}\right) \frac{C_{i+1,j+1} - C_{i+1,j-1}}{2\Delta x} + \frac{1}{2}\sigma^2 \frac{C_{i+1,j+1} + C_{i+1,j-1} - 2C_{i+1,j}}{\Delta x^2} = rC_{i,j}.$$

- Combine like terms (合并同类项):

$$C_{i,j} = a^* C_{i+1,j-1} + b^* C_{i+1,j} + c^* C_{i+1,j+1}.$$

Explanations and Improvements

- By the recursion above, one can obtain all the initial values $C(0, x)$, including the option price.
- Similar to a trinomial tree (三叉树) algorithm, but the coefficients (系数) a^* , b^* , and c^* can be negative.
- When the coefficient can be negative, the convergence does not always hold. This makes unstable results and huge errors.
- Improvements: implicit difference (forward recursion), hopscotch (跳房子) method, Crank-Nicolson method, etc. See John Hull's book.

Monti-Carlo (蒙特卡洛) Simulation

- Principle: use the sample mean to approximate the expectation:

$$\frac{1}{N} \sum_{i=1}^N x_i \sim E[X].$$

- The famous Buffon's needle problem (蒲丰投针实验) is an application of Monti-Carlo Simulation. It records the frequency of dropping needles crossing the grid on a paper, to approximate the circular constant π .
- Explain the option price as risk-neutral expectation, which enable one to apply Monti-Carlo Simulation to approximate option prices.

Example: European Option under the BSM

- Risk-neutral pricing:

$$C = e^{-rT} E_Q[(S_T - K)^+],$$

where

$$S_T = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T} Z}.$$

- Generate N standard normal samples z_i . MATLAB can output them directly. In C++, one can make it via applying central limit theorem or right continuous inversion on uniform random samples.

$$C \approx \frac{1}{N} e^{-rT} \sum_{i=1}^N \left(S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma \sqrt{T} z_i} - K \right)^+.$$

Sample Variance

- Sample variance:

$$S^2(N) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x}(N))^2.$$

- The $(1 - \alpha)$ confidence interval is

$$\left[\bar{x}(N) - z_{\alpha/2} \frac{S(N)}{\sqrt{N}}, \bar{x}(N) + z_{\alpha/2} \frac{S(N)}{\sqrt{N}} \right].$$

Therefore the order of convergence is $\frac{1}{\sqrt{N}}$.

- Note that $z_{0.025} = 1.96$, hence we often roughly judge that the true value is within 2 times the standard error.

Variance Reduction Technique

- Principle: consider another random variable S , which is positively correlated with C (the aim of simulation).
- S also partially captures the deviation of C from $E[C]$.
- “Hedge” the error.
- Option price and its underlying asset are often highly correlated. So we usually adopt $\bar{S}$ to reduce the variance of $\bar{C}$:

$$\bar{C}_{cv}(b) := \bar{C} + b(e^{rT} S_0 - S_T).$$

It is still an unbiased estimation (无偏估计) of C , since $E[S_T] = e^{rT} S_0$.

Minimize the Variance

- The new variance is

$$\text{Var}(\bar{C}_{cv}(b)) = \text{Var}(\bar{S})b^2 - 2\text{Cov}(\bar{S}, \bar{C})b + \text{Var}(\bar{C}).$$

- The value of b that minimize the formula above is

$$b^* = \frac{\sum_{i=1}^N (C_i - \bar{C})(S_i - \bar{S})}{\sum_{i=1}^N (S_i - \bar{S})^2} > 0,$$

which leads to

$$\text{Var}(\bar{C}_{cv}(b^*)) = \text{Var}(\bar{C}) - \frac{\text{Cov}^2(\bar{S}, \bar{C})}{\text{Var}(\bar{S})} < \text{Var}(\bar{C}).$$

An Experiment

A numerical example: $r = 0.06$, $\sigma = 0.35$, $K = 50$,
 $S_0 = 55$, and $T = 0.5$.

Number of Simulation Trials	1000
Simulated stock price	55.0083 +/- 0.433175
True call option value	8.97191
Simulate call price	8.98124 +/- 0.348425
Simulated call price with control	8.97485 +/- 0.105586
antithetic call price	9.06188 +/- 0.225048

General Diffusion Model: Euler Discretization

- One of the advantages of Monti-Carlo simulation: valid for all the derivatives and all the models.
- For general diffusion models, we can apply the Euler discretization to simulate many sample paths:

$$x(t_{i+1}) = x(t_i) + \mu(x(t_i), t_i)(t_{i+1} - t_i) + \sigma(x(t_i), t_i) \sqrt{t_{i+1} - t_i} Z_i.$$

- Of course, this introduces systematic errors, and makes the results to be biased.
- Still convergent

Comments

- Monti-Carlo simulation has wide applications. It is quite easy and straightforward to transplant, and is thus popular in the industry.
- Output the result as well as the standard deviation, which can indicate the accuracy in some ways.
- Slow convergence ($-1/2$ power), expensive computational cost
- As a result, one tends to apply other reliable methods (if any) rather than Monti-Carlo simulation.
- Suitable for benchmark (对照), since it always works.

Transform Inversion Method

- Principle: many quantities or functions in physics, engineering and finance proved to have no closed-form, but have integral transforms in closed-form.
- Theoretic tool: numerical integration and numerical inversion, e.g., fast Fourier transform (FFT).
- We take bilateral Laplace transform as an example, which nests Fourier transform and ordinary Laplace transform:

$$L_f(s) := \int_{-\infty}^{+\infty} e^{-st} f(t) dt.$$

The Inversion Algorithm

- Laplace inversion:

$$f(t) = \frac{1}{2\pi i} \lim_{T \rightarrow +\infty} \int_{\alpha - iT}^{\alpha + iT} e^{st} L_f(s) ds.$$

- Discretization and truncation (截断):

$$f(t) = \frac{1}{2\pi} \sum_{n=-N}^N e^{t(\alpha + inh)} L_f(\alpha + inh) h, \quad \text{where } h := \frac{T}{N}.$$

- Two sources of errors can be controlled by letting $h \rightarrow 0$ and $Nh \rightarrow +\infty$.

Application

- Laplace transform of European options:

$$L_C(s) = \frac{e^{-rt} S_0^{s+1}}{s(s+1)} L_{X_t}(-s-1),$$

where L_{X_t} is determined by the model.

- For exponential Lévy models, Laplace inversion of X_t is shown directed in its Laplace exponent:

$$L_{X_t}(s) = e^{G(-s)t} = E \left[e^{-sX_t} \right].$$

- Latest academic progress provides theoretical upper bounds of the errors, which can make the inversion results accurate enough within any arbitrary pre-specified error tolerance.

Multidimensional Application

- Terminal payoff of spread option is $(S_{1T} - S_{2T} - K)^+$, whose Laplace transform is given by

$$L_{\text{Spr}}(\vec{s}) = e^{-rT} K \frac{B(-s_2, s_1 + s_2 - 1)}{s_1(s_1 - 1)} L_f(-\vec{s}),$$

where B is Beta function.

- Laplace transform of barrier options under Kou's jump-diffusion model:

$$L_{\text{UIC}}(\vec{s}) = \frac{S_0^{s_2+1}}{s_2(s_2 + 1)(s_1 + r - G(s_2 + 1))},$$

where G is the Laplace exponent.

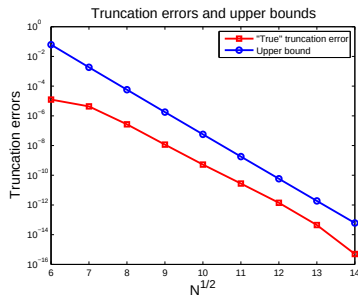
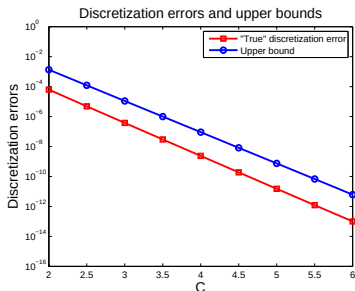
Spread Option Pricing under Variance-Gamma Model with Error Bounds

K	Cai & Shi	e_D Bounds	e_T Bounds	Hurd & Zhou	True Error
2.0	9.727458	6.7E-6	2.2E-5	9.727458	0.000000
2.2	9.630006	4.1E-6	1.2E-5	9.630005	0.000001
2.4	9.533200	2.6E-6	7.1E-6	9.533199	0.000001
2.6	9.437040	1.8E-6	4.3E-6	9.437040	0.000000
2.8	9.341528	1.2E-6	2.7E-6	9.341527	0.000001
3.0	9.246662	8.5E-7	1.8E-6	9.246662	0.000000
3.2	9.152445	6.1E-7	1.2E-6	9.152445	0.000000
3.4	9.058875	4.5E-7	8.2E-7	9.058875	0.000000
3.6	8.965954	3.3E-7	5.7E-7	8.965954	0.000000
3.8	8.873681	2.5E-7	4.1E-7	8.873681	0.000000
4.0	8.782057	2.0E-7	3.0E-7	8.782057	0.000000

Sources: Cai and Shi (2014, SSY)

Effect of Error Controlling

Introduce new parameters to make the two sources of errors approach 0.



Asymptotic Expansion Method

- Similar with transform inversion algorithm, asymptotic expansion (渐进展开) methods are also designed to compute the risk-neutral expectations in a roundabout (迂回) manner.
- Imagination: quickly compute the terminal net value (净值) of an investment with an annual return of 5% and a maturity of 10 years.
- The n -th order term of Taylor expansion involves n -th differentiation. Similarly, the n -th order term of Malliavin calculus involves n -fold Itô integral.

The Chaos Expansion

Wiener-Itô chaos expansion: for a quadratic integrable random variable ϕ , there exists such a unique sequence of deterministic quadratic integrable functions $\{f_n\}_{n \geq 0}$ where f_n is n -ary ($n\bar{\pi}$), that ϕ can be approximated by

$$\phi = \sum_{n=0}^{+\infty} I_n(f_n).$$

Here,

$$I_n(f_n) := n! \int_0^T \int_0^{t_n} \cdots \int_0^{t_3} \int_0^{t_2} f_n(t_1, t_2, \dots, t_n) dW_1 dW_2 \cdots dW_{t_{n-1}} dW_{t_n}.$$

Explanations

- The proof of Wiener-Itô chaos expansion is mainly based on the Martingale Representing Theorem.
- Example. $W_T \in \mathcal{F}_T$ is quadratic integrable. We have

$$W_T = T + I_2(1).$$

- In financial applications, we often calculate expansions in form, and then validate the convergence via the theory of Malliavin calculus.

Frontier Progress

- Pricing discretely monitored Asian options:

$$C(\Delta) := E_Q \left[e^{-rT} \left(\frac{1}{m+1} \sum_{j=0}^m S(j\Delta) - K \right)^+ \right].$$

- Cai, Li and Shi (2014, MOR) proves its expansion

$$C(\Delta) = \sum_{j=0}^J \Omega_j(z) \epsilon^j + O(\epsilon^{J+1})$$

to be convergent, where each term $\Omega_j(z)$ is a polynomial of normal random variables.

- Consequently, the expansion formula can be computed very quick, although very long.

Numerical Results (CEV model $\beta = -1/4$)

K	Malliavin	MC	Std. Err.	Abs. Err.
80	0.93506	0.93460	0.00054	0.00046
85	0.88533	0.88540	0.00073	-0.00007
90	0.80568	0.80598	0.00093	-0.00030
95	0.69779	0.69856	0.00109	-0.00077
100	0.57141	0.57236	0.00118	-0.00095
105	0.44099	0.44145	0.00119	-0.00046
110	0.32057	0.32173	0.00112	-0.00116
115	0.21979	0.21879	0.00100	0.00100
120	0.14247	0.14132	0.00085	0.00115

Effect of Convergence (under the BSM)

